

SOLUTION DESIGN DOCUMENT

Document Control

Document Metadata

Field	Value
Author(s)	 Person
Date	 Date
Status	[Draft / Under Review / Approved]
Target Audience	[Specify Target Audience]

Revision History

Version	Date	Author	Description of Change
0.1	 Date	 Person	Initial outline setup

1. Executive Summary & Scope Boundaries

1.1. Business Overview & Context

Document the business challenge, pain points in the current workflow, and the high-level business goals.

1.2. Scope Boundaries

Define explicit system boundaries (in scope & out of scope) to prevent scope creep during delivery.

1.3. Target Architecture Overview

Provide a high-level visual representation and description of the core system components, hosting environments, and orchestration layers.

[Insert High-Level System Architecture Diagrams Here]

1.4. Alternatives Considered

Compare chosen technical selections against viable alternatives, detailing trade-offs and rationale.

2. Production-Ready Future State Design

Provide details on future extensibility and architecture details of the solution as it would evolve in the future and as it will be rolled out into production.

3. System Flows, Sequence Diagrams & Agent Design

Sequence diagram outlining end-to-end data flow. Detail out the agent design and any pre-processing before agent invocation or optimizations for business requirements.

[Include Path Sequence Diagrams]

4. Security, Governance & Identity

Document authentication boundaries, network isolation, RBAC / identity management, Sensitive data handling & PII management.

5. Integration Details & Error Handling

Detailed methodology for 3rd party tool integration.

Map potential system component failures to custom fallback logic and user notifications.

6. Cost Estimation & FinOps

Key Cost Drivers: Identify primary variables impacting operational costs (e.g., token consumption, hosting tiers, search storage).

7. Deployment & Delivery Plan

Detail deployment plan / environments (Infrastructure as Code), state management, and configuration versioning.

Document the phased delivery milestones, dependencies, and deliverables.

8. Assumptions, Constraints, Risk & Mitigations

List all critical technical and operational assumptions regarding the target environment.

Identify key risks and define concrete mitigation strategies.

9. Quality Evaluation & UAT Framework

Define quantitative performance metrics, evaluation dataset curation, and acceptance thresholds.

10. Assumptions / Open Questions

Document the assumptions made, track outstanding design decisions, assigning ownership and deadlines.